

Presumed self-portrait, c. 1512.
Biblioteca Reale, Turin

Leonardo da Vinci

1452 to 1519

His notes on painting, gathered after his death into the *Trattato della pittura*

Leonardo advises the painter to learn many noses and mouths by heart, so that a new face can be set against ones he knows.



Isabella d'Este, c. 1499 to 1500.
Musée du Louvre

[. . .] impara prima a mente di molte teste, occhi, nasi, bocche, menti, gole, colli e spalle. [. . .] quando hai dato un'occhiata al volto della persona che vuoi ritrarre, guarderai poi in parte qual naso o bocca se gli assomigli [. . .]

[. . .] first learn by heart many heads, eyes, noses, mouths, chins, throats, necks and shoulders. [. . .] when you have cast a glance at the face of the person you want to portray, you will then look, in private, which nose or mouth it resembles [. . .]

Leonardo da Vinci, *Trattato della pittura*, part three, 286.

What distance takes away

As one moves away from a body, the outlines go first, then the smallest parts, and in the end only a mass remains.



Virgin and Child with Saint Anne, c. 1503 to 1519. Musée du Louvre

Le prime cose che si perdono nel discostarsi dai corpi ombrosi sono i termini loro; [. . .] e così successivamente si perdono le parti più minute, di modo che a lunga distanza solo rimane una massa ovale di confusa figura.

The first things that are lost as one moves away from shaded bodies are their outlines; [. . .] and so, in succession, the smallest parts are lost, so that at a great distance only an oval mass of confused shape remains.

Leonardo da Vinci, *Trattato della pittura*, part three, 422.

A lesson from the sculptor

A sculptor says that a mistake is hard to mend; Leonardo answers that a skilled hand takes away so little at a time that the work comes out right.



Studies of a horse, c. 1490.
Royal Collection, Windsor Castle

Potrà dire lo scultore, che dove fa un errore non gli è facile il racconciarlo; [. . .] Noi sappiamo bene che quello che sarà pratico non farà simili errori, anzi con buone regole andrà levando tanto poco per volta, che condurrà bene la sua opera.

The sculptor may say that where he makes an error it is not easy for him to mend it; [. . .] We know well that one who is skilled will not make such errors; rather, with good rules he will go removing so little at a time that he will bring his work well to its end.

Leonardo da Vinci, *Trattato della pittura*, part one, 34.

Subject

Where the lectures go

Entry

5



Match what you see against known parts.

Convolutional neural networks



The small parts fade first; the large shape stays.

U-Nets



Take away so little at a time that the work comes out well.

Diffusion